

Figure 5.8 Automated DCP Mounted on a Trailer

completed with generally a seating drop followed by three drops at different load levels (typically 9000, 12,000, and 16,000 pounds [40, 53, and 71 kiloNewtons] for highway testing). The pavement deflection under each drop is recorded automatically. The loading plate and sensors are then hydraulically lifted and the device is moved to the next test site.

Testing is usually accomplished in the driving lane at 100- to 500-foot (30–150 meters) intervals. During testing, the temperature of the pavement

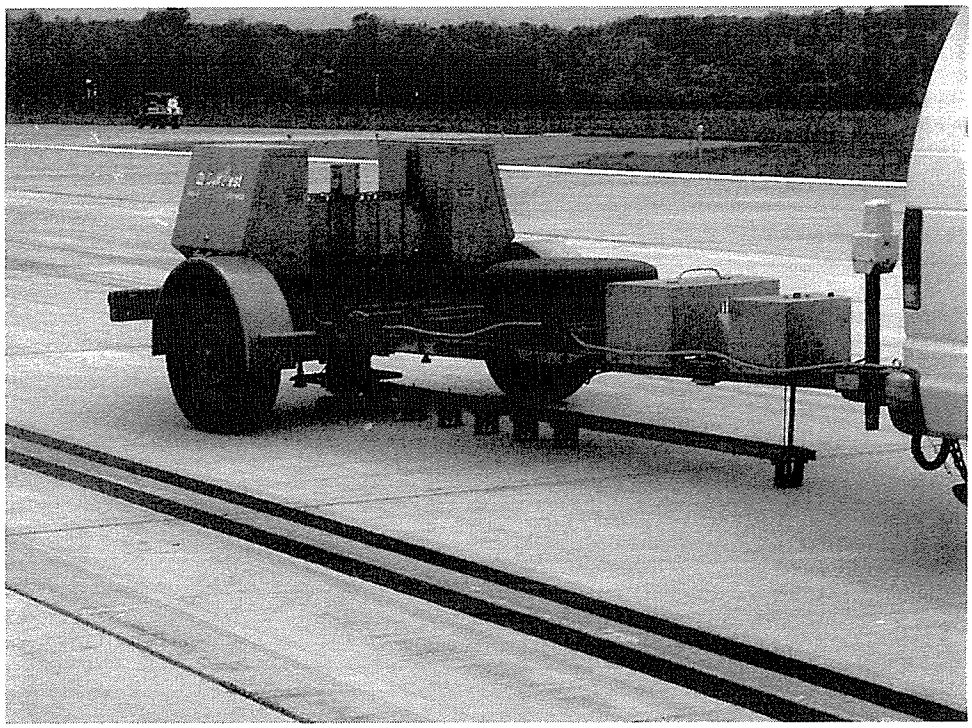


Figure 5.9 Falling Weight Deflectometer

should be monitored to allow corrections in the asphalt layer modulus due to temperature. As temperature increases, the asphalt modulus decreases, and as temperature decreases, the asphalt modulus increases. This correction is needed to ensure that the back-calculation process more accurately estimates the stiffness of base and subgrade layers. It is also necessary to determine layer thicknesses. Pavement deflection is sensitive to layer thickness, especially the thickness of the surface layer. If not available from the plans, the information can be obtained from cores or borehole logs.

Analysis of Data

The measured surface deflections are used as input to a back calculation procedure to estimate the stiffness of the pavement layers and underlying subgrade soil. The pavement is viewed as a multilayered elastic system, and deflections are measured at radial offset from the center of the load plate to define the deflection basin. The typical approach to back calculation for HMA pavements involves an iterative process in which an elastic layer program is used to determine theoretical deflections. In the first iteration, assumed layer moduli (i.e., "seed" moduli) are used to determine a theoretical deflection basin, which is then compared with the measured deflection basin. The moduli are then adjusted and the deflections recalculated until the calculated deflections match the measured deflections.

5.2.4 Strength Correlations

Most methods used for the design of highway and airfield pavements are based on elastic-layered theory. The preferred method for determining the strength of the subgrade soils and unbound subbase/base materials is the resilient modulus test. But the equipment and the test procedure is complex and time consuming. Therefore, a number of correlation equations have been developed to estimate the resilient modulus of unbound materials. Table 5.5 summarizes the correlations used.

Table 5.5
Correlations for the Determination of Resilient Modulus

Test Procedure	Equation (M_r in psi)
California Bearing Ratio (CBR)	$M_r = 2555 * (CBR)^{0.64}$
R-value	$M_r = 1155 + 555 * R$
Plasticity Index (PI) and percent of material passing the No. 200 (75 μm) sieve (w)	$CBR = \frac{75}{1+0.728*(w* PI)}$
Dynamic Cone Penetrometer (DCP)	$CBR = \frac{292}{DCP^{1.12}}$

5.3 Drainage of Asphalt Pavement Structures

5.3.1 Introduction

Adequate drainage of the pavement structure is considered the most important element in ensuring long-term pavement performance. The proper design of the drainage system for a particular pavement is of paramount importance. This section of the manual is a brief overview of the issues to be considered in designing pavement drainage. For additional details, consult MS-15, *Drainage of Asphalt Pavements* or the resources listed at the end of the chapter.

Reason for Drainage

Highway engineers recognize the critical need for good drainage in designing and constructing pavements. Probably no other feature is as important in determining the ability of a pavement to withstand the effects of weather and traffic, and in providing trouble-free service over long periods of time. Reducing the moisture content of the granular base and subgrade through the use of effective drainage directly affects fatigue cracking of the HMA surfaces, and thus the life of the pavement.

Water in a subgrade is usually slow to evaporate or drain. Even in dry weather, the subgrade soils may remain damp or wet indefinitely. Often, voids in porous materials become subterranean reservoirs unless they are drained.

With a few exceptions, most subgrade soils contain silt and clay, which weaken when wet. This weakening may be hastened by vibrations from traffic. If the subgrade is not properly shaped and sloped for drainage, water may collect at the interface of the granular base or subbase and the subgrade soils. As this water saturates the subgrade, the subgrade soils tend to lose stability and thus the strength required to support the loads on the pavement. Under traffic, wet soil may be worked into a fine-grained slurry that pushes up into voids in the granular base or subbase course, thus destroying the grain-to-grain interlock and weakening the granular materials.

Design

In developing the various features of a drainage system, consideration should be given to the following principal purposes:

1. To collect and drain water from the pavement surface
2. To collect subsurface water in the granular layers and keep the subgrade moisture content as low as possible
3. To prevent or retard embankment erosion
4. To intercept water from surrounding areas and carry it from the area
5. To lower the groundwater table

Drainage problems are caused by either surface flow or subsurface flow. Each requires separate analysis and treatment.

Surface drainage includes the disposal of all water present on the pavement surface and nearby ground. If possible, surface water should be prevented from entering the subdrainage system. Inevitably, some water enters the pavement structure, but may be kept to a minimum by surface maintenance procedures (e.g., crack sealing, etc.).

Subsurface drainage deals with water in the granular layers and in the subgrade soils beneath the pavement. Such water may enter the pavement through open joints, cracks, and other openings in the surface, through adjacent porous material, or from the ground below.

Subsurface water may be present in several forms, including these:

- Water that is free to flow under the force of gravity, referred to as "free water"
- Water that moves by capillary forces, referred to as "capillary water"

Important differences exist between surface and subsurface drainage. Surface runoff is usually a direct result of rainfall or melted snow. Water flow to a subdrain depends entirely on the permeability of the soil and the pavement courses through which the water must pass. The maximum flow of surface water into a subdrain usually occurs within hours of a rainfall event. In contrast, the maximum flow into a subdrain from groundwater sources may occur several hours or even days after rainfall events. Design requirements in the first case are based on the amount of rain or melted snow and, in the second case, on the permeability of the soil or aggregate.

5.3.2 Surface Drainage

Design for Roadways

Protecting pavement from the action of surface water is most easily accomplished with full-width asphalt paving. This means that the traffic lanes and the shoulders are paved with HMA to provide an impervious cover that prevents surface water from directly entering the pavement structure during rainfall. The pavement surface should be free from holes and cracks, have a permanently tight joint with the shoulder or gutter, and be shaped and sloped (crowned) for adequate surface runoff. Surface water usually is collected at the edge of a paved surface in ditches or gutters, which discharge the runoff by gravity flow to a location remote from the pavement. To prevent erosion on embankments, water should be directed to spillways by way of curbs or dikes constructed on the extreme edge of the shoulder. In addition, if either the volume or the velocity of the flowing water is expected to be excessive, these features can be constructed using asphalt lining, or other impermeable materials, to reduce the potential for erosion.

For urban sections, the paved surface, including shoulders, is crowned and typically drained by a curb and gutter system leading to catch basins, which carry the surface flow to a system of storm drains. If asphalt gutters are used, it is usually possible to tack the surfaces to ensure that no leakage occurs at the joint between the shoulder and the gutter/curb. Where concrete curb and gutter is used, it is critical to ensure that the joint between the paved surface and the concrete curb is maintained over the life of the pavement because the two materials respond to temperature in different manners and thus the joint tends to open.

In addition, the sidewalk or sodded area beside the curb must be graded to drain into the gutter to prevent water ponding in this location. Ponded water on the backside of the curb will seep into and saturate the subgrade at the curbline and may cause increased localized stress to the HMA layers.

Highway pavements are also built with a crown sloped to facilitate the flow of water from their surfaces into ditches at each side. The most common design practice allows the surface water to flow across the shoulder into a ditch or gutter at the side. On a pavement uniformly sloped to either the outside or the inside (such as a superelevated curve), the upper and lower shoulder crossfall is generally the same as that of the driving lanes to address safety concerns.

Paved shoulders provide the most effective surface drainage solution. From a drainage point of view, a less-effective practice (but still acceptable) is to provide gravel shoulders with increased crossfall to promote surface flow and minimize seepage into the subgrade. Sodded shoulders tend to prevent erosion, but they may also prevent water from draining freely from the road surface, thereby increasing the amount of moisture entering the ditch and saturating the subgrade soils at the edges of the pavement. Because sodded surfaces are typically weaker than granular surfaces, they may create a traffic hazard if they are used adjacent to higher-speed traffic lanes.

In some cases—where profile grade is sufficient—a raised lip or curb 4 inches (100 millimeters) or more in height may be constructed at the outside shoulder edge to intercept water and guide it to some convenient outlet. In this design, the edge of the pavement is sometimes shaped into a shallow channel which leads into storm sewer openings or other paved channels. Curbs and dikes should be located so that they do not interfere with the removal of snow from the pavement surface.

Drainage ditches receive water from pavement surfaces and subsurface drains. When the surrounding area slopes toward the roadway, these ditches also intercept water that would otherwise reach the roadbed. It is usually preferable to widen rather than deepen a drainage ditch to increase its capacity. Side slopes usually vary from about 4 (horizontal): 1 (vertical) on the roadway side to about 8:1 or flatter for the outside slope. The ditch should be large enough to accommodate the estimated surface runoff. Generally, the ditch should be constructed on a continuous smooth grade without depressions or pockets that will collect water.

Water collected in ditches may discharge directly into natural channels or into storm sewer openings. In other cases, it may be directed from the roadway by special ditches constructed for that purpose.

The ideal velocity of flow in a ditch is one that is fast enough to prevent settling of suspended material but not so fast as to cause erosion or scouring. Generally, a figure of 2 feet per second (0.6 meters per second) for newly seeded loam or fine sand, to 6 feet per second (1.8 meter per second) for well sodded or coarse gravel linings is an acceptable range of safe velocities. Higher velocities require measures such as rock dams, rock lining, or paving to prevent erosion. Hot mix asphalt is a desirable material for ditch paving.

5.3.3 Design for Large Paved Areas

Surface drainage design procedures for parking lots are the same as those for airports. A recommended guide is the Federal Aviation Administration's Advisory Circular 150/5320-5B, *Airport Drainage.*

After the parking area plan has been established and an efficient parking pattern worked out, drainage grades that obviate patrons stepping from cars into gutters are designed.

To prevent puddles from forming in rainy weather, ideally a minimum of 2 percent, or ¼ inch per foot (20 millimeters/meter), grade is necessary. On sloping terrain, constructing a grade presents no problem; but in flat areas, built-in drainage valleys give rise to a gently undulating flow on the paved area. Flows to proposed inlets should be calculated for existing runoff conditions, and if necessary, spacing should be adjusted to give satisfactory results. For average conditions, inlet spacing of 200–400 feet (60–120 meters) is satisfactory.

Because only a fixed contour can be laid in one pass of a paving machine, designs requiring a meandering gutter or valley should be avoided. Such gutters require hand placing, the quality of which is inferior to machine laying. A layout should provide for drains that follow straight lines. When the mix is being placed by the paving machine, the direction of the paver should parallel these lines. For the most efficient paver use, these lines should parallel the long dimensions of the lot. Watertight longitudinal joints are essential to prohibit the intrusion of surface water and to minimize the possibility of erosion under traffic.

In some instances, local bylaws restrict the flow of water into the storm sewers during rainfall events. In these situations, the pavement surface is sometimes used for short-term storage (tens of minutes) of precipitation. In these conditions, the pavement surface should be adequately maintained to prevent saturation of the pavement structure during the temporary storage period.

In order to drain the granular materials and subgrade, the use of stub drains 15–20 feet (5–7 meters) in length installed in a cruciform pattern at each catch basin is recommended. The stub drains should be installed in trenches with an invert below the underside of the granular layers and

should be connected to permanent year-round frost-free drainage, such as the catch basin system.

5.3.4 Subsurface Drainage

Introduction

Subsurface water results from rainfall or melting snow soaking into the soil and percolating downward until an impervious layer is reached. At this point, the water saturates the void spaces of the soil. The boundary between the saturated and unsaturated soil is referred to as the groundwater table. Subsurface water tends to be in motion and will percolate long distances through a porous layer if confined by impervious layers or strata.

A porous or cracked pavement may permit rainwater or melted snow to enter the granular base or subbase and saturate the various layers in the pavement structure. This water may then flow downgrade through the granular layers and create problems in other areas. These problems may be avoided most effectively with proper attention to pavement mix design and construction. A dense asphalt-treated base is not as subject to weakening and may be particularly effective. However, water infiltration into the subgrade soils is still a concern. The installation of subgrade drainage is fundamental to long-term pavement performance. If the structural section contains open-graded layers, these should be drained by pipes or other positive means to prevent water ponding.

Subsurface seepage, under certain conditions, may develop a hydrostatic head sufficient to lift the pavement completely off the base, causing cracking, and in extreme cases, complete disintegration of the pavement structure. The problem is more acute when steep grades in high rainfall areas are involved. With steep gradients, the water may travel longitudinally in an untreated base course along the direction of the gradient and cause excessive hydrostatic pressures at sag vertical curves or on the low side of superelevated horizontal curves.

Most urban pavements would benefit from the installation of subdrains along both curblines. The subdrains should be installed in trenches with an invert below the underside of the granular layers and should be connected to permanent year-round frost-free drainage such as the catch basin system. When full-length subdrains cannot be installed, the use of stub drains 15–20 feet (5–7 meters) in length on the upstream side of the catch basins is recommended.

In many cut sections, longitudinal subdrains are needed to prevent water from higher ground accumulating under the pavement. If not intercepted, subsurface water from a cut may flow onto an adjacent fill section and cause slumping of the fill slope and cracking of the pavement. The choice of filter material and the design of the drainage system must be given careful attention, considering both the type of material to be drained and the quantity of water to be expected.

Filter Material and Drain Rock

The purpose of pavement subdrainage is to enhance the long term (10 to 15 years) performance of the pavement. Thus, it is necessary to design the drainage system for long-term performance. Such a system requires the use of filters around drainage pipes to ensure performance. The filter allows water to flow to the drain and, at the same time, to prevent the loss of the fine particles from the soil. The design of a granular filter for any particular soil was defined by Terzaghi and Peck as follows:

$$D15, \text{filter} < 5 * D85, \text{soil (for soil retention)}$$

$$D15, \text{filter} > 4 * D15, \text{soil (for adequate permeability)}$$

where D_{xx} refers to the diameter of the soil or filter particle at the stated percent passing.

The material that meets these criteria for most subgrade soils is washed mortar sand, which is a clean, essentially one-size sand. Unfortunately, the permeability of mortar sand, while greater than that of most subgrade soils by several orders of magnitude, is still low for high-flow situations. Consequently, for adequate drainage, a perforated pipe is typically required. Such pipes are usually wrapped in a geotextile sock to prevent the sand particles from washing into the pipe.

Drain rock, on the other hand, essentially consists of one-size stone material (i.e., greater than ¼ inch [4.75 millimeters]), which results in very high permeability. It is apparent, however, that drain rock will never meet the retention criteria for most subgrade soils. As a result, the fine particles from the adjacent subgrade soils will wash into the drain stone and clog the material. This loss of soil will result in settlement above and/or beside the subdrain.

The permeability of a granular filter material is reduced very rapidly as the percentage of fine material is increased. To prevent these issues, a filter fabric is sometimes placed between the drain stone and the subgrade soils. The choice of geotextile and the proper installation are critical to the success of this type of construction. An inappropriate geotextile will still allow fines from the subgrade to pass and result in clogging and settlement.

Subsurface Drainage Schemes

The following approaches to subsurface drainage are suggested.

As noted previously, the installation of continuous subdrains under each curb in pavements designed with an urban cross section will extend the life of the pavement. If the expense of a continuous system is prohibitive, a system of stub drains could be used. For large paved surfaces, this same stub drain system installed in a cruciform pattern at the catch basins is also recommended.

For rural cross sections (e.g., shoulders and side ditches), adequate depth of the ditches is critical. The invert of the trench must be below the base of the pavement structure and typically at least 3 feet (1 meter) below centerline grade to promote adequate drainage. In addition, if granular layers are used in the pavement structure, these layers should be daylighted (i.e., placed full

width and exposed on the face of the ditch) to promote drainage. "Bathtub" construction, where the granular materials are only placed beneath the HMA surface, should be avoided as any water that enters the pavement structure will have difficulty escaping, even if cross-drains are placed at the profile sags.

Subsurface water can collect at the toe of cuts if appropriate drainage is not provided. If the material encountered at grade is stratified, it will be necessary to construct the drain to intercept the permeable layers and to remove seepage water. A weakness of this type of construction, however, is that water frequently bypasses the drain and rises to the surface beneath the pavement. This is caused by variations in the local permeability of the soils.

If an appreciable gradient is involved, a cross-drain should be placed at the downhill end of the cut. This feature will intercept any water flowing longitudinally that, if not drained, could saturate the fills and cause slumping of the fill slopes.

A properly designed, compacted, and adequately maintained dense-graded HMA pavement will be impermeable to water. However, in some localized conditions, subdrainage may still be required.

Drainage Related to Frost Action

When the frost depth exceeds the thickness of the pavement structure, a subsurface drainage system may be required to lower the water table to prevent the formation of ice lenses and the damaging effects of frost heaves.

Frost heaves are a serious problem. It should be noted that all saturated or partially saturated soils will expand in volume (heave) when the water in the pores expands by 9 percent on freezing. However, because the water content of most subgrade soils is less than 15 percent, the overall expansion is less than 2 percent. This likely applies to most subgrade soils in North America. True frost heave, from a geotechnical perspective, only occurs in a particular type of soil where the D_{50} of the soil is in the range of 30–60 microns and the uniformity coefficient (D_{60}/D_{10}) is less than about 6. This is a medium-size uniform silt. This type of soil has sufficient permeability to attract water in the capillary zone to the freezing front and is also fine enough to have a high capillary rise. This results in ice lenses forming in the soil. The ice lenses grow in a downward direction (as water in a pond freezes, being fed by water from below), as seen on Figure 5.10. In order for ice lenses to grow in the soil, there must be freezing soil temperatures, a water table close to the frost line, a soil favorable to the rapid movement of capillary water, and a soil capable of retaining water. When freezing temperatures are prolonged and the frost line penetrates susceptible soil, the ice lenses that form cause a heave at the surface.

To avoid damage to drainage structures, determining the depth of frost penetration is necessary for design purposes. The most reliable information is obtained from local records of depth of frost penetration in the area and for the soils concerned. When such local information is not available, frost penetration can be estimated from charts obtained from the National Oceanic and Atmospheric Administration National Climatic Data Center, (www.ncdc.noaa.gov).

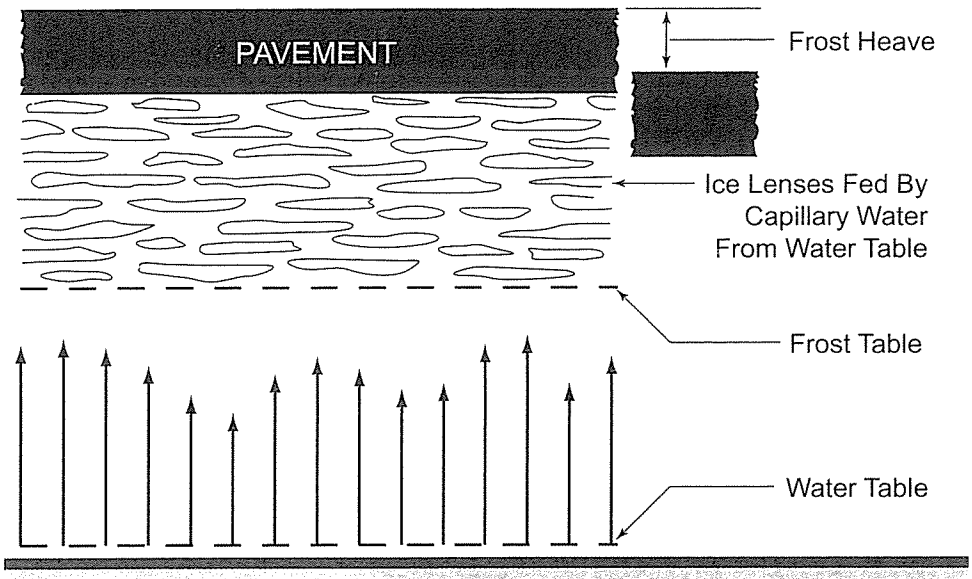


Figure 5.10 Ice Lenses Forming Beneath the Pavement Structure Cause Frost Heave

Problem soils should be removed to a depth equal to at least half, and preferably three-quarters, of the normal depth of frost penetration for the area of concern, and replaced with soil not susceptible to frost heave.

An alternate, but less effective, method of construction is to remove, mix, and recompact a suitable depth of subgrade to promote uniformity. This method will eliminate differential heave, although the whole road still will rise and fall. Even if every care is taken, a generalized form of heave can still occur. Frequently, ice accumulates between the granular base and the asphalt layer. This is probably caused by the migration of water vapor from the warm areas deep in the structure toward the colder surface layers. The sudden change in permeability at the asphalt layer restricts the flow of vapor causing it to condense and freeze.

Another aspect of frost susceptibility is thaw weakening of subgrade soils. This behavior is exhibited by most subgrade soils when the amount of material finer than no. 200 sieve (75 microns) is greater than about 30 percent. Because the majority of subgrade soils are affected, thaw weakening is generally the most serious aspect of frost susceptibility. The greatest loss of subgrade support is encountered when a wet autumn is followed by a winter with many freeze-thaw cycles and then by a rapid spring thaw. As frost advances into the soil, the available water in the voids freezes and the saturation of the soils reduces; as a result, water from below the freezing front is attracted to the freezing front and accumulates. In the spring, the frozen subgrade thaws from both above and below. The result is a period where the additional water drawn into the subgrade soils is liquid but trapped over an impermeable frozen zone. The abnormally elevated moisture content significantly weakens the soil. Soil strength will only increase after the frost is completely out of the subgrade and normal downward drainage occurs. The effect of thaw weakening can be reduced

by installing subgrade drains that have an invert depth below the frost line so that they provide drainage year-round. Over time, these drains will reduce the moisture content of the subgrade, which results in less accumulation of water as the freezing front advances, and thus reduce the thaw weakening.

An additional adverse effect of the spring thaw may occur in granular bases over fine-grained subgrades. The fines can be carried into the granular base, transforming it from a satisfactory base material into one of low load-carrying capacity. Under the right conditions a sand filter blanket, approximately 6 inches (150 millimeters) thick, or a filter fabric, can prevent the migration of fines.

The effects of spring thaw can be minimized by using adequate depths of granular soils or full-depth asphalt pavement and by maintaining proper drainage.

5.3.5 Geosynthetic Fabrics

The design of subsurface drainage systems should give consideration to geotextiles as alternative designs. Geosynthetic fabrics (geotextiles) are manufactured from a number of different materials, including polypropylene, polyester, nylon, polyethylene, and polyvinyl chloride. The function of geotextiles, when used as filters, is to allow groundwater removal without the buildup of excessive seepage forces or water pressures. The geotextile must also prevent piping or subsurface erosion of the soil which leads to settlement. In these applications, water flows through the geotextile into a water-conducting medium, usually a trench filled with a selected drainage aggregate and a slotted or perforated pipe, which quickly removes the water.

The use of geotextiles for subsurface drainage may be categorized in accordance with the following three general applications.

- Geotextiles are used as an envelope surrounding the single-size, clean coarse aggregate (i.e. less than 20 percent retained on a No. 4 sieve (4.75 millimeters)) in a trench. The geotextile is intended to prevent subgrade soil from moving into and eventually clogging the stone while still allowing water to pass through the geotextile. A more efficient system is to install drainpipes in the aggregate-filled trench rather than relying solely on the porous aggregate to carry the runoff. Typically, a 4- to 8-inch (100- to 200-millimeter) perforated flexible plastic pipe is installed within 5 feet (1.5 meters) of the surface. The sketch on the right in Figure 5.11 shows this configuration.
- A geotextile sock can also be used as a filter around a drainpipe, particularly for subsurface drainage in sandy or gravelly soils. However, in the presence of subgrade soil with more than 20 to 30 percent silt or clay (most subgrade soils), this type of subdrain tends to allow fines to migrate into the drain stone and to pass through the geotextile and clog the pipe, thus preventing drainage and also leading to

settlement of the pipe and/or road. If the trench is not lined with a geotextile, clean mortar sand surrounding a perforated pipe encased in a suitable geotextile sock (to keep the sand out of the pipe) has provided excellent long-term performance as a drain. The sketch on the left in Figure 5.11 shows this configuration.

- Geotextiles have successfully served as replacements to aggregate filters that prevent the migration of fines into untreated aggregate layers. In this type of installation, the geotextile typically lies between the layer with fines (subbase or subgrade) and a clean aggregate base layer. It should be noted that a geotextile will not work as a separation layer in cases of 'reversing' flow. This can occur with thin granular layers or with heavy wheel loads. The use of a separating layer also prevents the buildup of hydrostatic or seepage pressures.

Installation of geotextiles as filter layers is critical to performance. Sufficient care should be taken to ensure adequate overlap in both the vertical and horizontal directions and that the geotextile is installed according to manufacturer's recommendations. If the geotextile has gaps or tears, or if it has been installed with insufficient overlap, it will not perform as intended. Installation is relatively easy and cost-effective as compared to placing a two layer aggregate filter (a coarser aggregate next to the pipe and a finer filter aggregate next to the subgrade soils) in separate layers in a trench without contamination. Care must be taken in backfilling the trenches to ensure that the pipe is not crushed by construction equipment.

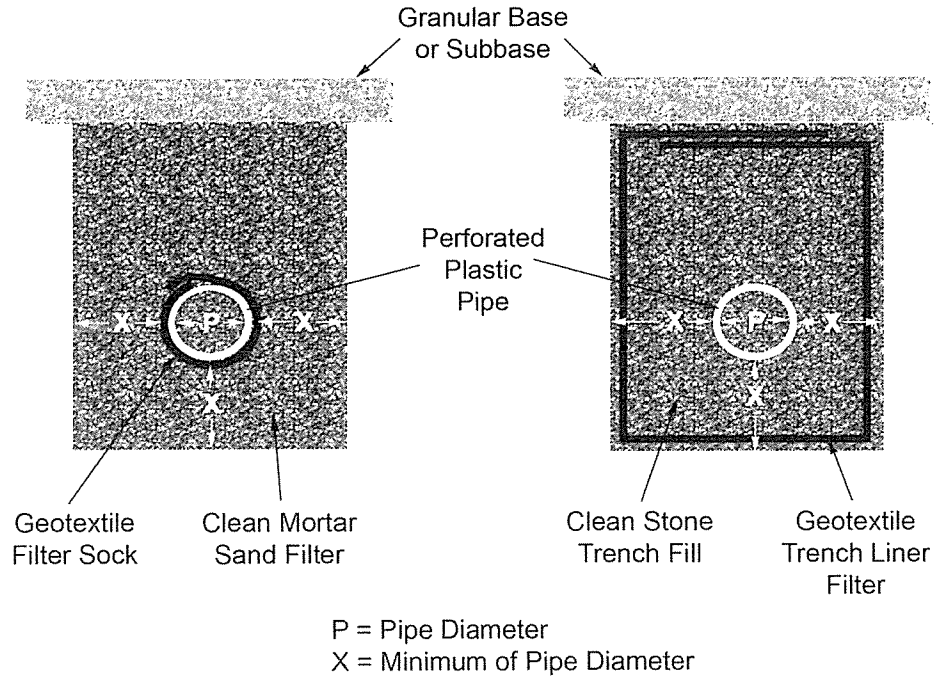


Figure 5.11 Two Typical Subdrain Installations

During design, it is important to ensure that the appropriate geotextile is selected for the particular application. More details on the use of geosynthetic fabrics are available from the manufacturers.

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CHAPTER 6

THICKNESS DESIGN

This chapter describes the Asphalt Institute thickness design methods and relates these Asphalt Institute methods to other mechanistic-empirical design systems. Readers are referred to other resources for additional information about these methods.

Section 6.1 Introduction

6.1.1 Design Principles
6.1.2 Input Variables
6.1.3 Highways, Roadways, and Streets
6.1.4 Airports
6.1.5 Heavy Wheel Loads
6.1.6 Planned Stage Construction
6.1.7 Overlay Design

Section 6.2 Relationship to Other Design Methods

6.2.1 Empirical Methods
6.2.2 Mechanistic-Empirical Methods
6.2.3 The Perpetual Pavement Method

Section 6.3 Design Manuals and Software

6.3.1 SW-1 Asphalt Pavement Thickness Design Software

6.1 Introduction

Thickness design for flexible pavements is an effort to economize the amount and the type of material used to support the projected traffic, on the given subgrade, under the specific climate conditions at the site. Whether thickness design is applied to major interstate highways, city streets, airports, or county roads, its objective is to strike a balance between two competing goals—structural capacity and life-cycle cost.

The state-of-the-art method in flexible pavement design is the Mechanistic-Empirical (M-E) method. The M-E method typically models the pavement as a multilayered elastic system and uses the principles of mechanics to calculate deflections, stresses, and strains at any point in the structure in response to an external wheel load. Calculated strains are then compared with observed pavement performance to determine the required pavement thickness. In the case of flexible pavements, two performance criteria are widely considered critical: limiting horizontal tensile strain at the bottom of the hot mix asphalt (HMA) layer to avoid fatigue, and limiting vertical compressive strain at the top of the subgrade to avoid rutting.

The Asphalt Institute method encompasses two types of design procedures, both of which have been incorporated into SW-1, *Asphalt Pavement Thickness Design Software.*

- **DAMA—*Advanced Structural Analysis.*** DAMA is a powerful computer-based M-E program that allows engineers to analyze any pavement system having up to five layers in which the materials can be described in terms of elastic modulus and Poisson's ratio. Engineers using DAMA consider monthly temperature variation, including freeze/thaw, and its effect on pavement materials to estimate remaining life to failure. DAMA is embedded into SW-1, *Asphalt Pavement Thickness Design Software.*
- **Design Curves.** The Asphalt Institute publishes pavement design curves in the following manuals for specific transportation design applications:
  - IS-154 *General Aviation Airports*
  - MS-1 *Highways, Roads, and Streets*
  - MS-11 *Air-Carrier Airports*
  - MS-23 *Heavy Wheel Loads* (Ports/Industrial Applications)

The curves presented in these publications were produced by presolving a large number of problems over a wide range of input values using DAMA. The resulting design curves provide users with a powerful solution based on M-E, in a handy manual that can be used in the office or the field.

In order to provide a single computer-based pavement design tool, the Asphalt Institute developed SW-1, a Windows-based program that offers a comprehensive suite of design solutions for highways, streets, airports, and heavy wheel loads, based on the design curves together with an improved

version of the DAMA advanced structural analysis routine. See Section 6.3.1 for additional information about SW-1.

The remainder of this chapter describes the Asphalt Institute design methods in more detail, relates the Asphalt Institute methods to other M-E systems, and refers readers to more detailed information about these methods.

6.1.1 Design Principles

The following paragraphs describe the design principles that are at the core of the Asphalt Institute methods. These principles are coded into DAMA and are inherent in the various design curves and SW-1.

DAMA models the pavement as a multilayered elastic system with up to five layers, including the subgrade (see Figure 6.1). The material in each layer is characterized by two variables—a modulus of elasticity and a Poisson's ratio. All layers in the model are assumed to extend infinitely outward in the horizontal plane. The subgrade layer is the lowest layer in the model and is assumed to extend infinitely downward (vertically). A condition of full-friction is assumed to exist at all layer interfaces.

A proposed pavement structure is subjected to traffic loading expressed in terms of the number of repetitions of a wheel load on either a single tire or a set of dual tires.

Temperature variability is known to affect the dynamic modulus of asphalt-bound materials and the resilient modulus of granular base and subgrade soils. DAMA allows for monthly variation of the mean monthly air temperature.

The response of the pavement system to the vertical wheel load, in terms of stresses, strains, and deflections, is calculated at a point in the structure using the principles of mechanics. Two strains are believed to be critical for design purposes:

- The **horizontal tensile strain** on the underside of the lowest asphalt-bound layer (either asphalt concrete or emulsified asphalt base). If the horizontal tensile strain at the bottom of the lowest HMA layer is excessive, fatigue cracking will be initiated.
- The **vertical compressive strain** at the top of the subgrade layer. Excessive vertical strain on top of the subgrade layer will result in permanent deformation (rutting) at the pavement surface.

Strain can be calculated, in both critical positions, as a response to repetitive wheel loads.

A cumulative damage calculation is used to determine the remaining life of the pavement structure. Miner's Rule is used to evaluate the ratio of the number of *actual* load repetitions to the number of *allowable* load repetitions. Miner's Rule states that the total allowable damage is equal to the summation of damage done by individual load applications. When the number of actual load repetitions is equal to the number of allowable load repetitions, cumulative damage is calculated to be 1, and a state of failure exists. At any level of actual load repetitions, cumulative damage can

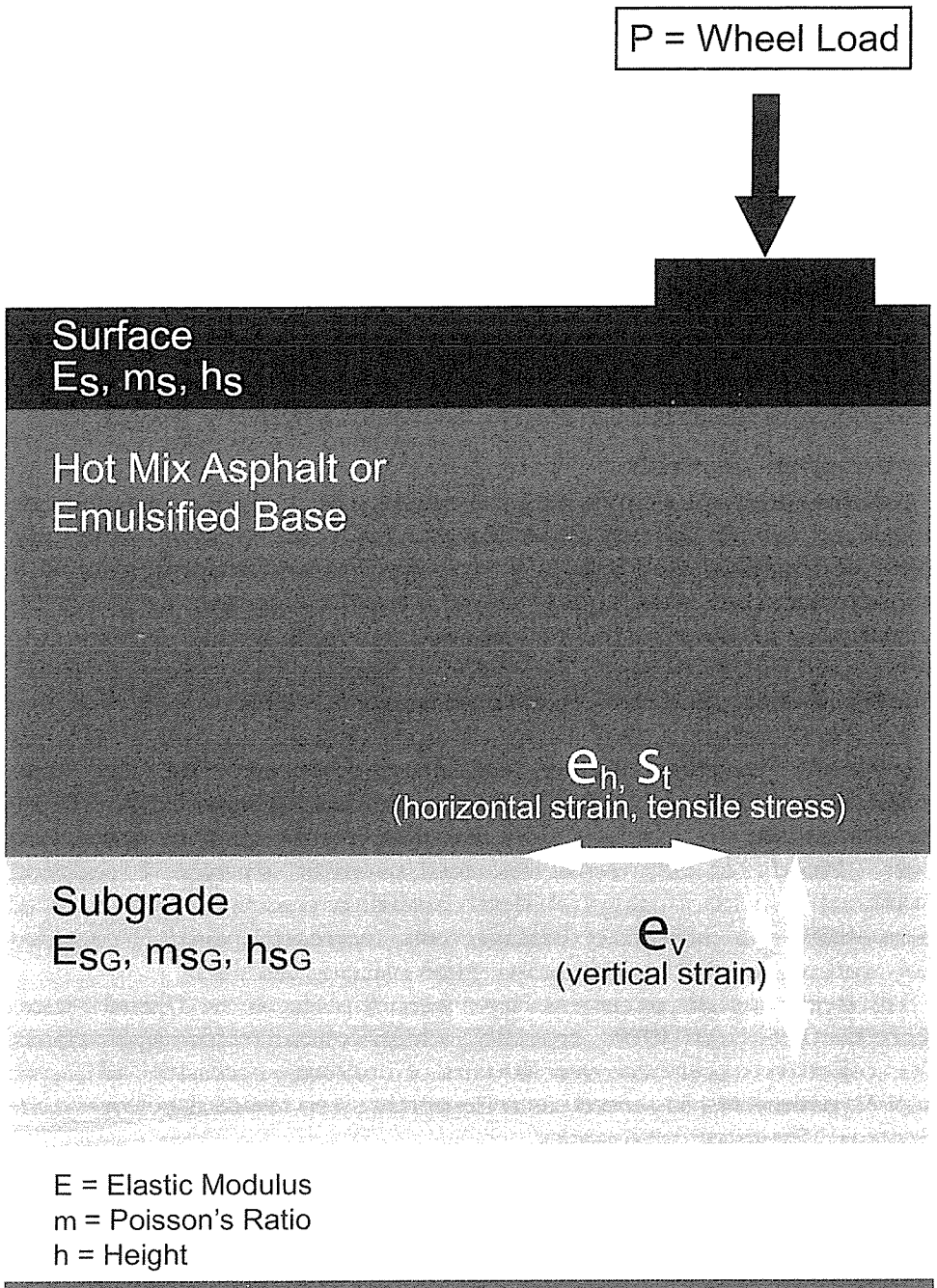


Figure 6.1 Layered Elastic Model

be evaluated for either condition of failure—horizontal tensile strain or vertical compressive strain.

Conventional HMA

Conventional hot mix asphalt pavement refers to the manner in which materials are typically incorporated into a flexible pavement structure.

Economics demands that pavement designers use lower-cost unbound granular base materials when possible to reduce the initial cost of the pavement structure. But while these materials are less expensive to produce and place, they are not as effective in distributing traffic-induced stresses to the subgrade. As a result, thick layers are required to support even low-volume truck traffic. Typically, conventional HMA pavements consist of thick aggregate base layers (6–30 inches) (150–750 millimeters) coupled with relatively thin HMA surface courses (3–5 inches) (75–125 millimeters). In a conventional design, the HMA wearing surface acts to provide a smooth riding, all-weather, waterproof surface, while the majority of the structural capacity is derived from the granular base layer. For many agencies in the United States, conventional HMA pavement sections provide an affordable solution for low- to moderate-volume roads.

Full-Depth Asphalt

A full-depth asphalt pavement is one in which asphalt mixtures are employed for all courses above the subgrade or improved subgrade. HMA or asphalt-treated base layers are placed directly on the subgrade. Asphalt-stabilized base layers in a full-depth pavement need not be constructed as thick as the comparable aggregate base layer in a conventional HMA pavement. The HMA wearing course creates a strong, durable, smooth, and cost-effective pavement structure.

Full-depth asphalt pavements are able to resist traffic loads in an efficient "low-profile" pavement section. Asphalt-aggregate mixtures resist these traffic-induced stresses far better than unbonded aggregate layers, which have no tensile strength. Therefore, an HMA layer spreads the wheel load over a broad area. As a result, less total pavement structure is required. Additional benefits of using full-depth asphalt pavements include reduced permeability and subsurface drainage costs, increased speed of construction, and an extended paving season when placing thick lifts.

Full-depth asphalt pavements have been in place in the United States since the 1960s and 1970s, especially in high-volume traffic applications. Our collective experience over 40 years of building successful full-depth asphalt pavements has served as an important step toward developing the Perpetual Pavement philosophy.

Perpetual Pavement

Perpetual Pavements are designed and built to last longer than 50 years without requiring major structural rehabilitation or reconstruction. Like full-depth asphalt pavements, Perpetual Pavements are designed using HMA or asphalt-stabilized layers in all layers above the subgrade. What sets Perpetual Pavements apart is the additional care taken in mix type and material selection at all layers within the pavement structure.

The Perpetual Pavement concept improves upon the full-depth asphalt design method by incorporating a fatigue-resistant base layer and rut-resistant intermediate and surface layers into the pavement structure. These mix type selection techniques will minimize fatigue and rutting in the pavement structure.

Perpetual Pavement design also requires careful attention to material selection. Hard angular aggregates must be carefully selected, processed, and blended together to form a strong "stone-skeleton" to support intended traffic loads. Properly selected Performance Graded asphalt binders will improve rut-resistance at high temperatures and diminish the effect of transverse low temperature cracking. The combined result of these mix type and material selection strategies is a pavement structure that requires only periodic surface renewal in response to distresses confined to the wearing course.

It should be noted that the Perpetual Pavement design concept is separate and distinct from the Asphalt Institute thickness design method. More detail on this method is provided in Section 6.2.3.

6.1.2 Input Variables

Data-input requirements for the Asphalt Institute design method depends on the procedure selected by the user. The design curves were produced and intended for everyday use and require minimal information as input into the procedure. DAMA, on the other hand, is intended for advanced structural analysis of pavement structures with the requirement of more-detailed user inputs.

With either procedure, there are four types of inputs required to use either AI method: climate, traffic, subgrade strength, and material properties. The following is a brief description of these input variables.

Climate

Air temperature is the climate-related user input in the AI method. Air temperature affects the design of the pavement structure in two important ways. First, the stiffness of HMA and other asphalt-stabilized layers is directly dependent on the pavement temperature, which can be calculated from air temperature. For example, high pavement temperatures result in a reduction of the stiffness of HMA layers. As layers with lower stiffness do not distribute loads as effectively as stiff layers, pavements at higher temperature are more susceptible to plastic deformation caused by concentration of load than pavements that are at low temperature.

For granular base and subgrade soils, temperature determines whether freezing and thawing conditions will occur. Frozen materials become stiff, while thawing weakens unbound materials, particularly subgrade soils.

In DAMA, mean *monthly* air temperature is used to directly characterize the climatic variation for a specific pavement structure using location-specific temperature data. For design-curve analyses such as MS-1, the user is asked to select a mean *annual* air temperature appropriate for a geographical location.

Traffic

The AI method is based on an analysis of individual wheel loads and the response of the pavement system to the wheel loads.

Traffic input in DAMA is a single or dual wheel load. The basic user inputs are the magnitude of the load, the contact pressure of the tire, and the number of monthly load repetitions. For dual tires, wheel-to-wheel spacing is also required. Its ability to analyze single or dual wheels allows DAMA to be applicable to virtually any pavement section being considered.

The design curves published in MS-1 are specific to highway vehicles. Rather than analyzing individual wheel loads, the curves have been developed based on the Equivalent Single-Axle Load (ESAL) concept. An ESAL is a single axle carrying an 18,000-pound load distributed to the pavement. Trucks, cars, and other highway vehicles with multiple axles and a broad range of loaded weights can be equated to a corresponding number of ESALs. By predicting the total number of ESALs expected over the design life of the pavement, an engineer can enter the design curves to calculate the required pavement thickness.

For air-carrier airport design MS-11, *Thickness Design—Asphalt Pavements for Air Carrier Airports*, traffic is expressed as the number of departures of a specific aircraft type. The design curves used for port and industrial applications in MS-23, *Thickness Design—Asphalt Pavements for Heavy Wheel Loads*, are based on a single-wheel analysis but allow the user to convert a multiple-wheel vehicle into an equivalent single-wheel load. Industrial pavements must be designed with knowledge of the specific vehicle loads and wheel geometry.

Subgrade Strength

Subgrade soils form the foundation of pavement structures. Some knowledge of a soil's suitability to resist deformation under loading is required to complete a pavement design.

The basis for subgrade-strength measurement in the Asphalt Institute method is the resilient modulus, M_r. This applies for both DAMA analysis and for solutions using the design curves.

In order to facilitate the use of other widely used field and laboratory tests, correlations have been established between resilient modulus and the California Bearing Ratio (CBR) and the resistance (R) value. Procedures for performing CBR and R-value tests are detailed in MS-10, *Soils Manual for the Design of Asphalt Pavement Structures.*

Material Properties

As stated earlier, DAMA allows up to five layers in the pavement section, each represented by an elastic modulus and a Poisson's ratio. Because Poisson's ratio is not expected to vary for a layer, the emphasis should be on properly characterizing a layer's elastic modulus.

When working in the DAMA advanced structural-analysis mode, the user can characterize the elastic modulus of a layer by a number of means. For asphalt-bound layers, the dynamic modulus can be entered directly for each month in the year or the user can use a modulus-temperature relationship to calculate the dynamic modulus based on the monthly average temperature. For specialized situations, mix volumetric properties can be prescribed to further refine the modulus-temperature relationship.

These three characterization methods provide the user with a great deal of control over the analysis.

For granular layers, the DAMA user is offered two methods of characterizing resilient modulus. The user can enter the resilient modulus directly for each month of the year. Alternatively, the user can prescribe a stress-dependent relationship that better models frost effects or moisture variations.

The subgrade layer allows for direct entry of the resilient modulus for each month of the year. This monthly variation allows the user to prescribe up to 12 data points per year, adequate for description of a freeze-thaw cycle.

The material property handling within the DAMA model provides the user with great influence to shape the model to best reflect actual conditions in the field.

For the pavement design curves published in MS-1, *Thickness Design—Asphalt Pavements for Highways and Streets,* MS-11, and MS-23, certain simplifying assumptions have been made regarding the material properties in the pavement structure. MS-11 and MS-23 design curves for air-carrier airports and heavy wheel loads are limited to full-depth asphalt pavements, where asphalt-bound layers are used for all layers above the subgrade. For MS-1, several commonly used pavement sections were modeled: full-depth asphalt, HMA-over-granular base, and HMA-over-emulsified base. For pavement sections that depart from these standard sections, the MS-1 design curves are not applicable.

6.1.3 Highways, Roadways, and Streets

The Asphalt Institute method of thickness design for pavements supporting highway vehicles is covered in detail in MS-1, and is one of the methods included in SW-1.

Engineers accomplish thickness design by graphical solution of the appropriate pavement design curve. Selection of the appropriate design curve is based on climate and anticipated material type (full-depth asphalt, asphalt-over-granular base, etc). These curves represent hundreds of presolved design problems over a wide range of input values, using the DAMA analysis tool. The result is an efficient design tool that is applicable to everything from interstate highways to residential streets and all roadway classifications in between. If the pavement supports highway-type vehicles, from automobiles to long-haul tractor semi-trailer vehicles (Figure. 6.2), then the MS-1 design curves apply. This method has also been incorporated into SW-1.

In order to simplify climate inputs for the MS-1 method, users are asked to select among three mean annual air temperature (MAAT) levels: 45°F, 60°F, and 75°F (7°C, 15.5°C, and 24°C). These MAAT levels bracket those found throughout much of North America and are representative of the climates in New York, South Carolina, and Arizona, respectively. Engineers should evaluate project-specific weather conditions to ascertain which of the three MAAT levels is appropriate for a particular project. It is possible to

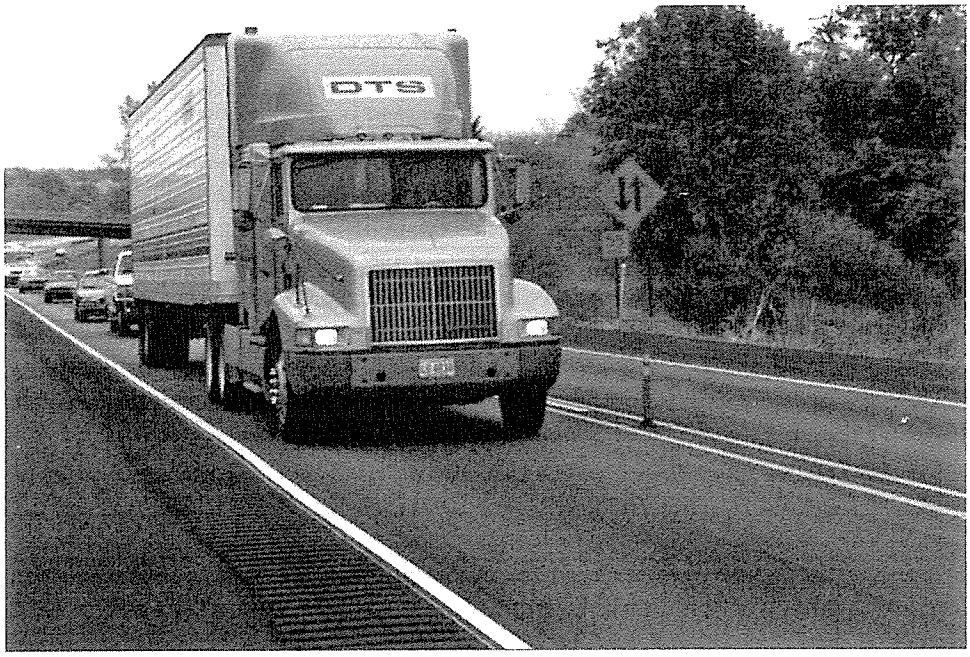


Figure 6.2 Typical Tractor-Trailer Used on Highways

interpolate between the results from two design curves to complete a design for MAAT levels that do not match one of the three standard levels.

Traffic analysis in this method involves estimating the number of ESALs to be supported over the design life of the pavement. Every highway-type vehicle, from small automobiles on two axles to large multiple-axle tractor-trailer combination trucks weighing more than 100,000 pounds (45,000 kilograms), can be measured in terms of ESALs. The process of equating anticipated vehicle traffic to ESALs is covered in detail in MS-1. SW-1 goes even further by automating ESAL calculations. Both methods allow the user to factor in traffic growth over the pavement design life.

Subgrade strength is characterized as resilient modulus, M_r. Field-test results such as CBR and R-value can be correlated to M_r using standard correlations. It is strongly recommended that subgrade conditions over the length of a project are carefully evaluated and that an adequate number of tests are used to establish a representative subgrade M_r. Additional guidance on soil testing can be found in Chapter 5 of this manual and in MS-10.

The MS-1 design curves were developed for the following six common highway and street design strategies:

- Full-depth asphalt concrete
- HMA-over-emulsified-asphalt base—Type I
- HMA-over-emulsified-asphalt base—Type II
- HMA-over-emulsified-asphalt base—Type III
- HMA-over-6-inches (150 millimeters) untreated aggregate base
- HMA-over-12-inches (300 millimeters) untreated aggregate base

The pavement design procedure for highways and streets allows engineers to consider a range of six material types and three MAAT levels. Designers only need to select the appropriate curve from the 18 design curves that are available based on climate and material type. For the selected curve, the designer enters the curve with input values for design ESALs and subgrade M_r. The intersection of these input values on the design curve produces the solution to the problem. SW-1 steps the designer through the process in a user-friendly interface and prompts the designer for the necessary information.

The SW-1 also allows a conventional design with aggregate base thicknesses varying between 4 and 18 inches.

6.1.4 Airports

Air-carrier airport pavements support heavy commercial jets with gross loads well in excess of 60,000 pounds (27,000 kilograms). A vast secondary network of thousands of smaller airports allows the nation's fleet of general aviation (GA) aircraft to serve local communities. GA aircraft have gross weights up to 60,000 pounds (27,000 kilograms) (Figure 6.3).

General Aviation Airports

The Asphalt Institute method of thickness design for GA airports is covered in detail in IS-154, *Thickness Design—Asphalt Pavements for General Aviation,* and is one of the methods included in SW-1.



Figure 6.3 Typical General Aviation Type Aircraft